

Subject Description Form

Subject Code	COMP5541
Subject Title	Machine Learning and Data Analytics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: 1. present the principles, concepts and models of modern machine learning; 2. introduce analytics skills to analyze data and get insight from it.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand the essential concepts of machine learning models and algorithms. b. develop insights into the effectiveness of machine learning techniques. c. gain knowledge of applying machine learning techniques to various cutting-edge applications. d. design machine learning solutions to solve new challenging problems in practice by considering different requirements and issues. e. participate in team work, presentation and technical writing.
Subject Synopsis/ Indicative Syllabus	• Supervised Learning Definitions, training data, dimensionality of input space, supervised learning steps, choice of algorithms, bias-variance dilemma, noise, k-NN, linear discriminant analysis, naïve Bayes, neural networks, deep learning models, fine-tuning, evaluation measures. • Unsupervised Learning Data types, dimensionality reduction, feature selection, k-means, mixture models, contrastive learning, anomaly detection, neural network-based approaches, Hebbian learning, deep belief networks, self-organizing map, latent variable models. • Semi-supervised Learning Supervised learning and unlabeled training data, continuity and manifold assumptions, generative models, heuristic approaches and low-density separation. • Reinforcement Learning Definitions, algorithm for control learning, criterion of optimality, value function, directed policy search, deep reinforcement learning, apprenticeship learning. • Data Analytics Applications Data types and pattern discovery in data using machine learning, selective applications in topic modeling, genomics, prediction, etc., selected issues like scalability, interpretability, legal/social/ethical issues.

Teaching/Learning Methodology	Lectures teach students on the main concepts of the course, together with comprehensive examples, and class questions and answers for easy understanding. Tutorials and lab sessions help students to review the learned concepts, master the practical techniques and necessary tools for effective system/application development. Group project offers the opportunity to students to develop analytical and problem solving skills through system implementation and interpersonal communication.																																																		
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1" data-bbox="536 600 1481 1137"> <thead> <tr> <th data-bbox="536 600 842 801" rowspan="2">Specific assessment methods/tasks</th><th data-bbox="842 600 1002 801" rowspan="2">% weighting</th><th colspan="5" data-bbox="1002 600 1481 734">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th data-bbox="1002 734 1098 801">a</th><th data-bbox="1098 734 1193 801">b</th><th data-bbox="1193 734 1289 801">c</th><th data-bbox="1289 734 1385 801">d</th><th data-bbox="1385 734 1481 801">e</th></tr> </thead> <tbody> <tr> <td data-bbox="536 801 842 869">1. Assignment</td><td data-bbox="842 801 1002 1003" rowspan="3">30</td><td data-bbox="1002 801 1098 869">✓</td><td data-bbox="1098 801 1193 869">✓</td><td data-bbox="1193 801 1289 869">✓</td><td data-bbox="1289 801 1385 869">✓</td><td data-bbox="1385 801 1481 869"></td></tr> <tr> <td data-bbox="536 869 842 936">2. Project</td><td data-bbox="1002 869 1098 936">✓</td><td data-bbox="1098 869 1193 936">✓</td><td data-bbox="1193 869 1289 936">✓</td><td data-bbox="1289 869 1385 936">✓</td><td data-bbox="1385 869 1481 936">✓</td></tr> <tr> <td data-bbox="536 936 842 1003">3. Test</td><td data-bbox="1002 936 1098 1003">✓</td><td data-bbox="1098 936 1193 1003">✓</td><td data-bbox="1193 936 1289 1003">✓</td><td data-bbox="1289 936 1385 1003"></td><td data-bbox="1385 936 1481 1003"></td></tr> <tr> <td data-bbox="536 1003 842 1070">4. Examination</td><td data-bbox="842 1003 1002 1070">70</td><td data-bbox="1002 1003 1098 1070">✓</td><td data-bbox="1098 1003 1193 1070">✓</td><td data-bbox="1193 1003 1289 1070"></td><td data-bbox="1289 1003 1385 1070">✓</td><td data-bbox="1385 1003 1481 1070"></td></tr> <tr> <td data-bbox="536 1070 842 1137">Total</td><td data-bbox="842 1070 1002 1137">100</td><td colspan="5" data-bbox="1002 1070 1481 1137"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessments consist of project, assignments, lab exercises, and tests, which are designed to facilitate students to achieve intended learning outcomes. Lab exercise is designed to encourage students to acquire good understanding of the relevant knowledge, practice in order to enrich their hands-on experience with various software tools. The project is designed to enhance students' ability to acquire the understanding and using different knowledge, principles, techniques, tools to solve a real problem through team. Quizzes are to ensure the students understand the concepts. Examination will evaluate student's understanding and usage of machine learning and data analytics techniques.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignment	30	✓	✓	✓	✓		2. Project	✓	✓	✓	✓	✓	3. Test	✓	✓	✓			4. Examination	70	✓	✓		✓		Total	100					
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Student Study Effort Expected	Class contact:																																																		
	▪ Class activities					39 Hrs.																																													
	Other student study effort:																																																		
	▪ Assignments, Projects, tests, Examination					66 Hrs.																																													
	Total student study effort					105 Hrs.																																													

Reading List and References	1. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 2. Christopher M. Bishop, Hugh Bishop, Deep Learning Foundations and Concepts, Springer, 2023. 3. Ian Goodfellow, Deep Learning, MIT, 2016. 4. Zhi-Hua Zhou, Machine Learning, Springer Nature, 2021. 5. EMC Education Services (Editor), Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, Wiley, 2015. 6. Han, J., and Kamber, M., 2022, Data Mining: Concepts and Techniques, 4th Ed., Morgan Kaufmann, San Francisco, CA. 7. Jakub M. Tomczak, Deep Generative Modeling, Springer, 2024.
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